

LIVER (Surgery) — Revision Table

Bold = high-yield.

ANATOMY & ASSESSMENT

Topic	Key points
Anatomy	MHV divides R (60%)/L lobes; 8 segments; Cantlie's line (GB fossa→MHV) ; falciform = umbilical vein remnant; bare area posterior
Blood supply	Portal vein 80% + hepatic artery 20% ; 3 hepatic veins; LPV > RPV; only organ that regenerates
ALF	INR >1.5 + encephalopathy + <26 wk, no prior disease ; mortality 30–40%
Chronic signs	Spider naevi, palmar erythema, leukonychia, gynaecomastia; flapping tremor = best sign
Prognosis	CTP (A 10%, B 30%, C 75–80% surgical mortality) ; MELD = INR + bilirubin + creatinine
Investigation	US first-line ; CT (HCC arterial, haemangioma late venous); MRCP biliary; biopsy resectable lesion CONTRAINDICATED (seeding)

TRAUMA & PORTAL HYPERTENSION

Topic	Key points
Liver trauma	2nd to spleen; blood loss main complication; FAST ; manage by haemodynamic stability NOT grade — stable → conservative
Portal HTN	Mostly cirrhosis; presents decompensated (encephalopathy/ascites/varices)
Variceal bleed	Resuscitate + vit K/FFP + platelets <50k + terlipressin/octreotide/somatostatin + prophylactic antibiotics ; band ligation → balloon → TIPSS
TIPSS	Refractory portal HTN; encephalopathy 40%, stenosis 50%/yr; CI = portal vein occlusion
Ascites	SAAG >1.1 g/dL = portal hypertension

SPECIFIC DISEASES & INFECTIONS

Condition	Key points
PSC	Young men + UC ; ↑cholangiocarcinoma/GB cancer; transplant before malignancy
PBC	Females ; anti-mitochondrial Ab; pruritus; transplant
NAFLD/NASH	Most prevalent chronic liver disease (NAFLD 25%); 20% NASH → cirrhosis
Budd-Chiari	Hepatic outflow obstruction; young women ; triad pain + ascites + hepatomegaly ; hypercoagulable; anticoagulate
Ascending cholangitis	Charcot's triad (jaundice + rigors + RUQ) ; E. coli; antibiotics + urgent drainage (ERCP)
Pyogenic abscess	Klebsiella/E. coli/Strep milleri; biliary source 35%; metronidazole + cephalosporin
Amoebic abscess	E. histolytica; anchovy sauce pus ; metronidazole + diloxanide; drain if left lobe/>5cm/poor response
Hydatid / Fasciola	Albendazole (pre+post surgery) / triclabendazole

TUMOURS

Tumour	Key points
Hepatic adenoma	Women + OCP ; 10% → HCC ; excise >5cm
FNH	Middle-aged women; hepatocytes + Kupffer; no malignant potential, no treatment
Haemangioma	Most common benign (F:M 5:1); resect only if symptomatic
Cysts	Simple (5%); polycystic (PKD, 10% cerebral aneurysm); cystadenoma (premalignant)
HCC	Commonest primary; HBV (>50%)/HCV/cirrhosis/aflatoxin ; BCLC staging ; resectable 20–40%; recurrence 80%/5yr; transplant = Milan criteria (≤5cm or ≤3 nodules ≤3cm); PVE if small FLR
ICC	2nd to HCC; Clonorchis (SE Asia) ; aggressive, 30% resectable, avoid biopsy, 5-yr 3%
CRC mets	70% of CRC; resection only cure (50% 5yr) ; 20% candidates; R0 1cm margin, FLR 25% ; 5-FU/oxaliplatin + anti-VEGF/EGFR; liver-first approach